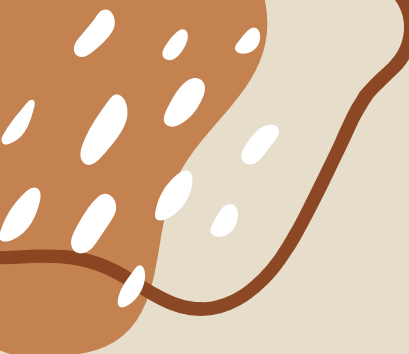




SCIENTIFIC PAPER TITLE GENERATION

Group 5

Vu Huu Thong

Le Duy Hai

Luu Quang Vu



TABLE OF CONTENTS

- 1. FORMULATE/OUTLINE THE PROBLEM**
- 2. IDENTIFY INPUTS AND OUTPUTS**
- 3. PREPARE DATA: INSPECTING & VISUALIZING**
- 4. OPTIMIZATION TECHNIQUES**
- 5. NEURAL NETWORK MODEL**
- 6. MEASURE PERFORMANCE**
- 7. INFERENCE ON NEW DATA**
- 8. CONCLUSION**

FORMULATE THE PROBLEM

Problem: Abstracts contain too much information and need to be condensed into a concise, succinct title.

Solution: Fine-tune the T5-base(Text-to-Text Transfer Transformer) model on the arXiv dataset to automatically generate accurate academic titles from paper abstracts.

Summary: The preceding three decades have seen the emergence, rise, and proliferation of machine learning (ML). From half-recognised beginnings in perceptrons, neural nets, and decision trees, algorithms that ...

Title: Machine Learning and the Future of Realism



IDENTIFY INPUTS AND OUTPUTS

INPUT: PAPER ABSTRACT

Long text detailing methods and results from scientific papers

OUTPUT: PAPER TITLE

Short text containing key keywords and main concepts





INPUT

```
Summary (first 200 chars): Because of their occasional need to return to shallow points in a search tree, existing backtracking methods can sometimes erase meaningful progress toward solving a search problem. In this paper, we ...
```

OUTPUT

```
Title: Dynamic Backtracking
```



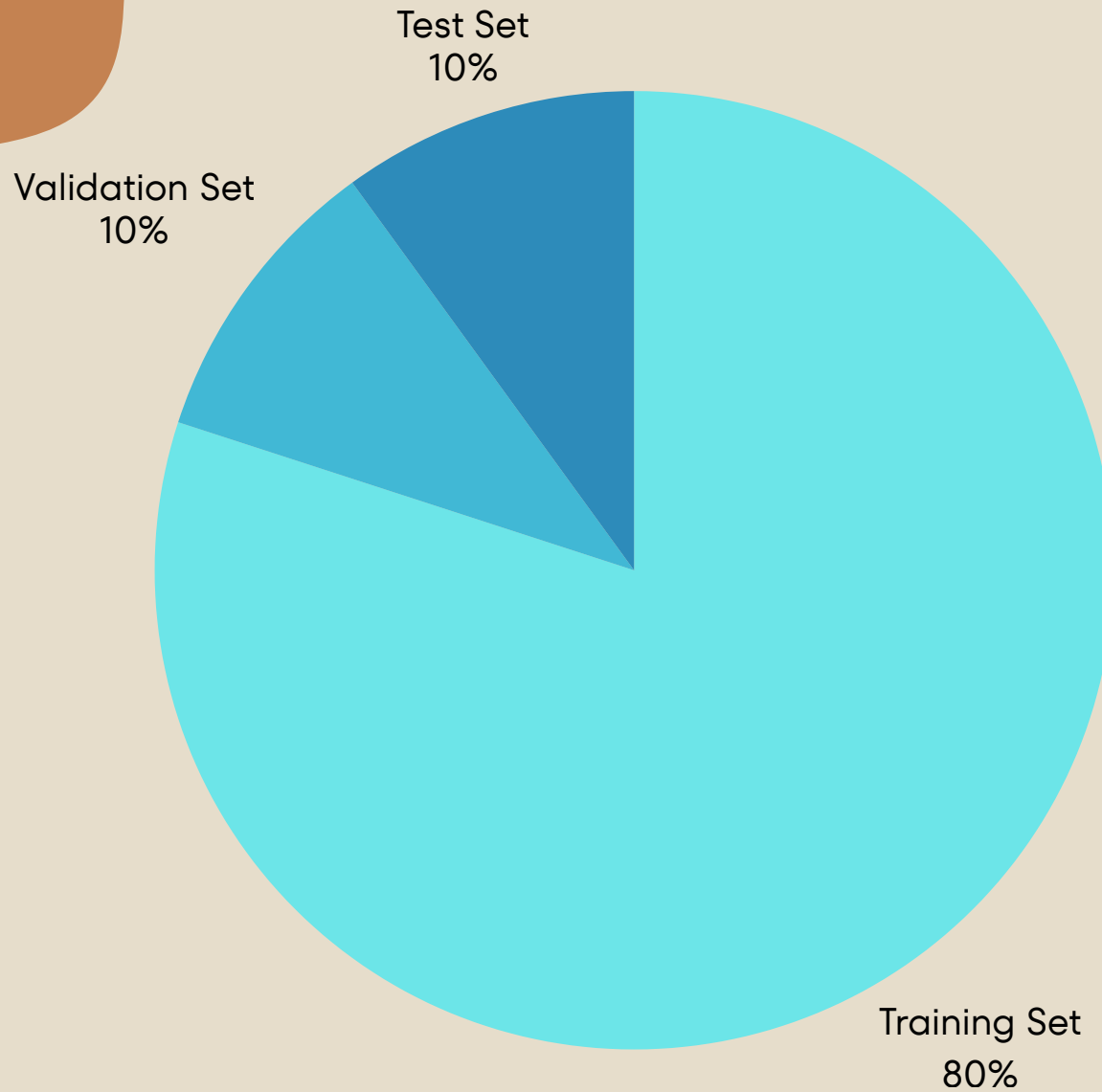
PREPARE DATA: INSPECTING & VISUALIZING

Dataset Overview

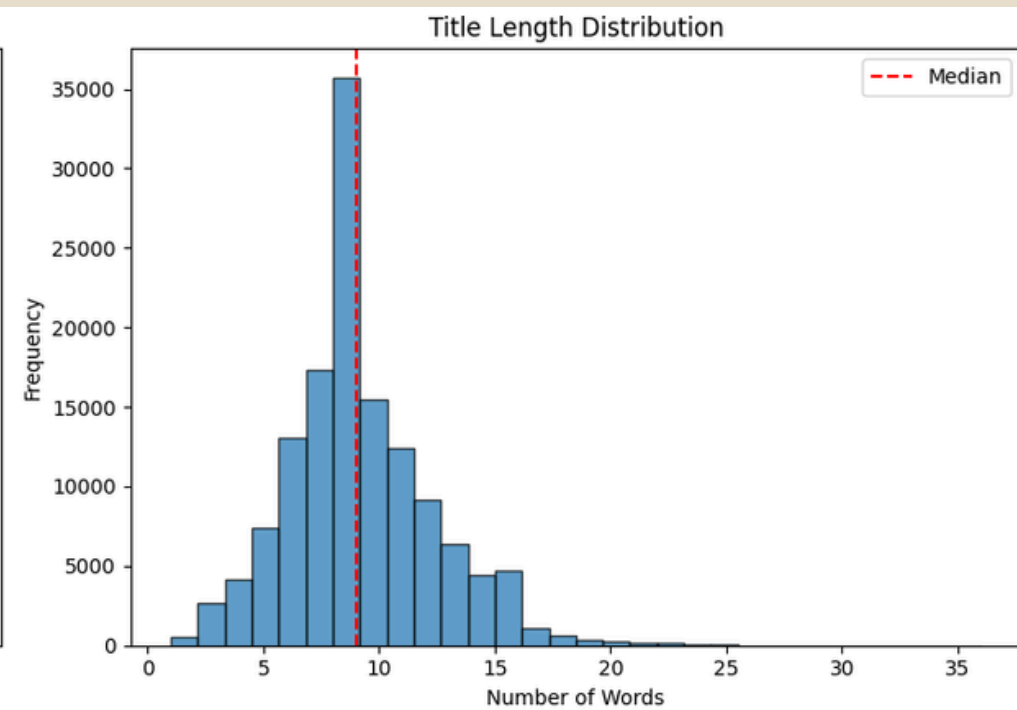
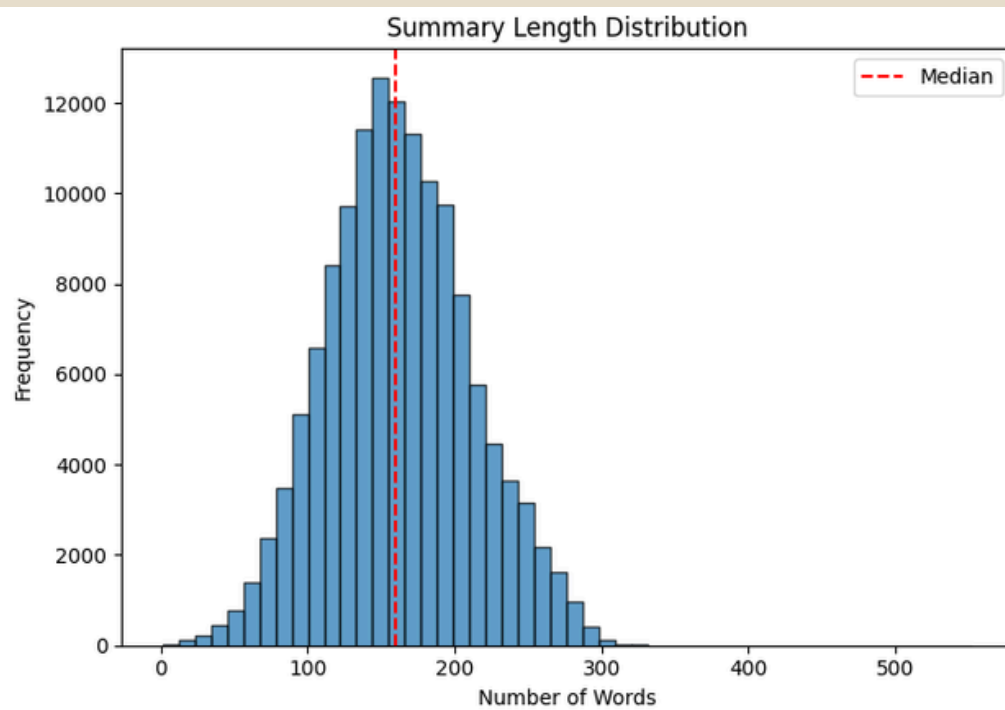
- **Total of 136,238 clean samples (after removing null values)**
- **Average summary length: ~162 words**
- **Average title length: ~9 words**

Data Statistics

Shows significant length differences between summaries and titles, highlighting the challenge of condensing information.



The chart above shows an allocation of 80% for the training set, and 10% for each of the validation and test sets.



OPTIMIZATION TECHNIQUES

Hardware Acceleration

Utilized Tesla T4 GPUs to accelerate computations and reduce training time

Text Preprocessing

Used AutoTokenizer for text encoding and added the "summarize: " prefix specific to the T5 model

Tokenization Workflow

Efficient batch processing with progress tracking to handle large-scale data from arXiv

Model Optimization

Fine-tuning strategies to maximize performance on limited GPU memory (15.83 GB available)

OPTIMIZATION TECHNIQUES

```
Using device: cuda
```

```
GPU: Tesla T4
```

```
Memory Available: 15.83 GB
```

```
Setup completed successfully!
```

```
Tokenizing datasets...
```

```
Tokenizing: 100%  23860/23860 [00:40<00:00, 534.77 examples/s]
```

```
Tokenizing: 100%  2982/2982 [00:06<00:00, 522.81 examples/s]
```

```
Tokenizing: 100%  2983/2983 [00:04<00:00, 697.29 examples/s]
```

```
Tokenization completed!
```

NEURAL NETWORK MODEL

- **Model:** AutoModelForSeq2SeqLM loaded from t5-base checkpoint
- **Architecture:** Encoder-Decoder mechanism converts Summary (Encoder) sequences into Title (Decoder) sequences
- **Model Size:** Approximately 892MB, containing millions of pre-trained parameters
- **Mechanism:** The Encoder processes input summaries, and the Decoder generates output titles token by token, leveraging the Transformer's Self-attention mechanism for semantic understanding

MEASURE PERFORMANCE

Model performance is measured by semantic similarity score

Analysis of low-match cases: Examining cases where the model's predictions do not match well helps to understand the model's limitations

Example of a low-match case:

- Reference headline: 'Strong Neural Network training...'
- Generated headline: 'Scale-invariant Adaptive Gradient Methods...'
- Similarity score: 0.28 (indicating the model generated a headline that was too long compared to the original)

Key Insights: Understanding failure cases helps identify areas for improvement and model refinement

MEASURE PERFORMANCE

Low Match Examples:

=====

Reference: Understand Functionality and Dimensionality of Vector Embeddings: the Distributional Hypothesis, the Pairwise Inner Product Loss

Generated: A Bias-Variance Trade-off for Dimensionality Selection for Vector Embeddings

Similarity: 0.21

Reference: Robust Training of Neural Networks Using Scale Invariant Architectures

Generated: Scale Invariant Adaptive Gradient Methods for Robust and Memory-Efficient Training of Large Language Models

Similarity: 0.28

INFERENCE ON NEW DATA

Interactive Functionality: Use the `generate_title` function with the `num_beams=4` parameter to find the best results through beam search.

Capability: The system can generate titles for any arbitrary summary text.

Demo Readiness: The interactive title generation function is ready to use with the command: `generate_title(your_summary, num_beams=4)`. This demonstrates the model's practical applicability for real-world use cases, allowing researchers to automatically generate accurate titles for their scientific papers.

INFERENCE ON NEW DATA

Example 2:

Summary: This paper introduces a new algorithm for image segmentation based on deep learning.
The propose...

Generated Titles:

1. Deep Learning Algorithm for Medical Image Segmentation
2. Deep Learning Algorithm for Image Segmentation
3. Deep Learning Algorithm for Medical Imaging Segmentation

CONCLUSION

Successfully built a complete NLP pipeline from data processing to model fine-tuning. The T5-small model operates efficiently on T4 GPUs with good performance

The model demonstrates strong capabilities in generating semantically relevant titles from summaries, with high accuracy on the test dataset

Expansion opportunities include using larger models (T5-base), training on more diverse datasets, and implementing advanced decoding strategies like Beam Search to improve title quality



THANK YOU

